P510/1

PHYSICS

Paper 1 (Theory)

3 hours

July/August 2026

NEPTA MOCK EXAMINATIONS – 2026

Uganda Advanced Certificate Of Education

PHYSICS

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(Theory)

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INSTRUCTIONS TO CANDIDATES

- This paper consists of two sections; **A** and **B**
- Section A consists of **two compulsory items**
- Section B consists of two Parts 1 and 2, each consisting of two items and you are required to **respond to any one item from each part**.
- Respond to a total of **four items** only.
- Any Additional Item(s) responded to will not be scored
- Start each item on a fresh page.
- Indicate the items you have answered in a grid on the front page
- Mathematical tables and silent non-programmable calculators may be used.

SECTION A

(Both items in this section are compulsory)

Item 1

Water performs a key role in running most activities in our homes, however, water scarcity is a general problem in most communities especially during dry seasons. A head of a family in one of the villages in Northern Uganda decided to purchase a large storage tank of capacity **10,000 litres** and of height **4m** in order to ensure that his family has enough water to use during long dry seasons. Before purchasing this tank, he made consultations from the family contractor who advised that if the tank is bought, it must be placed on a flat raised base.

The contractor informed the family members that he was to come on site for further guidance and its better he finds the tanks on site. This interaction with the contractor prompted the head of the family together with his elder son who is in S6 vacation and waiting to pursue a Bachelor's Degree in Civil engineering to drive the family truck to the nearby trading Centre to purchase the tank.

The road to the shop was generally flat but with a circular bend of radius **50m** banked at **30°**. Before setting off they wanted to know the maximum safe speed along the circular bend for the truck not to skid. After purchasing for the tank, the shop attendant referred them to a plumber who told them that he was to fix an outlet pipe to the tap at a height of **10.0cm** from the bottom of the tank which confused the family head why he chose this height. They also expressed interest in knowing the maximum speed at which water will be ejected through the pipe.

At the shop, a crane was used to load the tank on the family truck and they drove home. On their way back home, a fast – moving bus bypassed them and blew off the cover of the tank which puzzled the family head.

On reaching home, they were challenged with offloading the tank. The son proposed that a rough wooden plank of length **7.0m** could be used in offloading it at constant speed from the truck of height **1.5m.** The family head however suggested that lowering it vertically downwards could be a better option. During offloading using the wooden plank, a lot of noise was made which scared the family head but the son informed him that the origin of this noise can be explained and the noise can be minimized.

Hint:

- Acceleration due to gravity = **$9.81ms^{-2}$**
- Density of water = **$1000kgm^{-3}$**
- Total mass of the empty tank = **250kg**
- Coefficient of friction on the wooden plank is **0.12**
- Assume the banked road is frictionless

Task

(a) Analyze the information provided and use it to address the concerns raised.

(b) Explain any one real life application of the effect that blew the cover from the tank.

Item 2

Three university students, Peter, Paul and John are carrying out a project for a prototype T.V set to be made so that it is affordable for locals in Ugandan villages. One of the expensive parts of the set is the square fluorescent screen which should have a limited area of $\mathbf{1600}\boldsymbol{cm^2}$ so as to cut down on costs.

The screen will be placed **20.0cm** beyond the deflecting system which comprises of two pairs of deflecting plates, one of which is arranged vertically and the other horizontally. Each plate is **5.0cm** long and each pair of plates is separated by a distance of **2.0cm.**

Peter suggested that, since the Uganda domestic electricity is supplied at **240V,** this should be used as the tube voltage for the T.V set to avoid costs for step-up transformers, he further proposed a variable voltage of a maximum of **165V** to be connected across each of the deflecting plates.

Paul is worried about the size of the screens, and thinks that with a deflecting voltage of **165V** the images may turn out to be much bigger than the screen which will make viewing the T.V an inconvenience.
His other concern is that some life-threatening radiations may be produced during the use of the T.V sets. He doesn't know how such radiations are produced and how people can safe guard themselves from their effects.

John points out that, **90%** of the rural dwellers in Uganda may not be connected to the national power grid. However, he was told that the sun's radiation can be an alternative source to power the T.V sets, though he is not aware how this can be done.

Support material

Electron charge,	e	$= \mathbf{1.6 \times 10^{-19}}\boldsymbol{C}$
Electron mass,	M_e	$= \mathbf{9.11 \times 10^{-31}}\boldsymbol{kg}$
Planck's constant,	h	$= \mathbf{6.63 \times 10^{-34}}\boldsymbol{Js}$

Speed of light, C $= \mathbf{3.0 \times 10^{8} ms^{-1}}$
Radiations of frequency of at least $\mathbf{3 \times 10^{15} Hz}$ are dangerous to human life.

Task

(a) Analyze the information and use it to address Paul's concerns.
(b) As a Physics student, help John understand how the suggested alternative source powers the T.V sets.

SECTION B

PART 1

(Respond to only one item from this part)

Item 3

A school management committee is planning to construct a multi-purpose hall for academic, sports, and community activities. During one of the planning meetings, the project engineer recommended using clay roofing tiles but the committee members didn't know why it is suitable for keeping the hall cool. Before purchasing the roofing materials, however, the headteacher instructed that samples of different roofing materials should be obtained and their thermal conductivities experimentally determined so that the most suitable material can be selected.

The engineer also proposed purchasing a **300W solar panel** to provide electrical power. Some members of the committee wanted to know the required surface area of the solar panel if it operates at an efficiency of **22%** and receives solar radiation from the Sun, whose surface temperature is **5700 K**.

In addition, one committee member suggested installing a device that uses lenses in the hall for displaying presentations and other information. Before approving the proposal, the committee requested an explanation of how it works.

The sound Engineer further advised that the positioning of the loudspeakers should be carefully considered. He explained that if the speakers are poorly positioned, some areas of the hall may experience very loud sound while other areas receive weak sound, even though the speakers are connected to the same audio source which puzzled the committee members and requested for a scientific explanation

Support material

Radius of the sun = **7.0×10^{5} km**

Radius of earth's orbit about the sun = **1.5×10^{8} km**

Task

(a) Analyze the information and provide solutions to the committee.

(b) Explain how the suggested device works.

Item 4

A certain school was invited to participate in a science fare organized at the district. The school administration hired a bus to transport learners to the venue, however, being a dry season, the learners felt hot while on their way to the venue in the bus but when they checked the windows of the bus, they were still closed which left them wondering how the inside of the bus became hot. Their physics teacher told them that it could be due to green house effect which they didn't know and how it results into high temperature in the bus.

On their way they wanted to see some geographical features but they couldn't see them properly since they were very far. A certain S.6 student amongst them told them that they can model a device using two lenses of the same type but having different focal lengths and use it to view distant objects especially when in normal adjustment. However, they didn't know how it works.

After realizing that they were getting late for the fare, the bus driver increased the speed but the teacher told him that traffic officers charge for overspreading after determining the vehicle's speed from a distance using a device that uses electromagnetic radiations, which confused the driver how it works.

At the science fare, one of the invited schools presented about liquids that can be used as coolant in motorcycles engines, however the discussant emphasized that the selection of these liquids is based on their specific heat capacity which can be determined in the physics laboratory using various methods amongst which is continuous flow method. The discussant also stressed that the best coolant should have a specific heat capacity of **at least 3740$Jkg^{-1}K^{-1}$.**

When they went back to school, they wanted to test whether some of the liquids in the laboratory are suitable coolants. In their trials, a solid was heated to **80°C** and was quickly transferred into a copper calorimeter of mass **200g** containing **100g** of a liquid at **20°C**, after stirring, the final temperature of the mixture was recorded as **42°C**, though the students are challenged with how to make judgement from these results.

Hint

Heat capacity of the metal = **276 $\mathbf{JK^{-1}}$**

Specific heat capacity of copper = **400 $\mathbf{Jkg^{-1}K^{-1}}$**

Task

(a) Analyze the information and help the students address their challenges.

(b) Help the driver understand how the device works

PART 2

(Respond to only one item from this part)

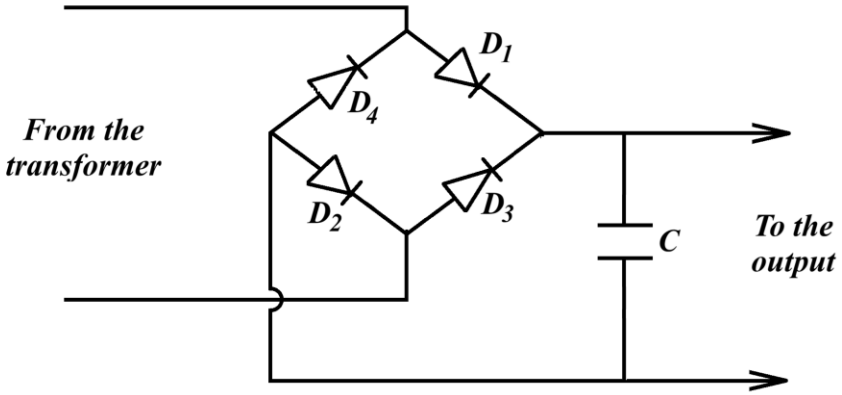
Item 5

Agriculture is one of the economic activities Ugandans engage in and the government emphasized the need for value addition on the agricultural products to make farming profitable. Mr. Jonah, a prominent fruit farmer in Kyazanga has established a fruit juice processing and packaging factory that uses an automated conveyor belt to transport packaged products using a D.C motor.

After several months of operation, the factory manager notices that the conveyor belt no longer moves at a uniform speed, instead, it repeatedly speeds up and slows down causing products to pile up at some points, creating delays in production and the motor becomes excessively hot during operation leading to increased electricity consumptions and frequent shutdowns.

The manager called a local mechanic as he was requested by the Mr. Jonah to help them solve the challenge. The mechanic took long to reach the factory and explained to Mr. Jonah that his car had a problem with its ignition system and he had to change its coil which took him a long time to get and fix. Mr. Jonah was interested in knowing how the whole system that ignites the car works because he had a similar challenge with his car.

On opening the motor's control circuit. The manager observed that the wires from the transformer were connected to a bridge of diodes and a capacitor before running to the motor as shown in the figure below.



The mechanic discovered that the insulating material in the capacitor of its smoothing circuit broke down and the manager asked how the capacitor enables the motor to run smoothly but the mechanic could not explain. The mechanic also found out that; the armature coil of the motor has a resistance of **5.2Ω** and when connected to **90V** supply, the current taken by the armature coil is **1.85A**. On contacting the engineer, he advised that the motor can still be used if its efficiency is above **90%** otherwise, they should install another motor. The manager needs information on how the insulator works in the capacitor and why the efficiency of the motor cannot reach **100%**.

The manager needed to compile a report to Mr. Jonah involving the role of the diodes in the circuit which the mechanic could not ably explain.

Task

(a) Analyze the information in the scenario to answer the manager's query and help him know whether they should install another motor.

(b) Help Mr. Jonah understand how the system works.

Item 6

An association of traders organizing a trade fare exhibition have received concerns from different committees that need to be addressed before the exhibition day.

In a preparation meeting, one trader alerted the organizing committee that some gangs always come with sharp metallic objects that can cause a security threat and insisted on the need to detect and arrest them at the entrance.

On consulting the head of security, they were told that a device can be installed to make an alarm whenever a metal is brought close to it but the organizers were not aware of how it works. The security officer added that the device is available though it has operational challenges that needed fixing.

The head of security took the device for repair and it was discovered that the coil of its circuit was faulty but they were unaware of its specification in order to replace it to enable the device work effectively when connected to an a.c source of frequency **90.9MHz** with its capacitor of capacitance **5.47pF** in the circuit unchanged.

James, one of the exhibitors needed a portable rechargeable speaker to use in advertising his merchandise on the exhibition day but did not have enough knowledge about it. He moved with his daughter, Emmanuella, an S.6 physics student to help in identifying the right specifications of the speaker that can work for **at least 2 hours** before running down once fully charged from a **100V** d.c supply. On reading the user manual of the available speaker, the following information was obtained.

- Power rating of the speaker was **1000W**
- Charge storage bank comprised of **two heavy duty capacitors** each of capacitance **0.36kF** connected in **parallel.**

James also wanted to know how the speaker can get fully charged from the d.c source available after it has run down.

Task

(a) Critically analyze the given information and advise the exhibition organizers accordingly.

(b) Help James to clear his queries.

END